Vebicorvir "Beats" Bersacapavir by Docking and Loses by 1.10 kcal/mol of Noise: A Vina Benchmark of HBV Capsid Assembly Modulators on 5WRE and the Functional-Cure Gap

AI-assisted draft — NOT peer-reviewed, NOT published. All claims require independent laboratory verification. Do not cite as established evidence.

Author: Zubair Ali, Pharm-D student, Shaheed Mohtarma Benazir Bhutto Medical University (SMBBMU), Larkana, Pakistan **Drafting assistant:** Lapis (AI) **Date:** 2026-11-07 · **Health Papers Agent — Paper #2** (daily 5-paper run)

Keywords: hepatitis B virus; capsid assembly modulator; bersacapavir; molecular docking; AutoDock Vina; core protein; functional cure; GLS4

Abstract

Chronic hepatitis B affected about 240 million people and caused about 1.1 million deaths in 2024, yet no approved drug clears the virus: nucleos(t)ide analogues suppress replication indefinitely and functional cure — sustained hepatitis B surface antigen (HBsAg) loss off treatment — remains rare. Capsid assembly modulators (CAMs) that bind the core protein dimer-dimer interface are the most advanced new mechanism, but no CAM is approved, and the most advanced, bersacapavir, reached phase 2b without a single functional cure in the REEF-2 trial. We benchmarked four CAMs against the same HBV core-protein pocket with AutoDock Vina 1.2.7 on PDB 5WRE (1.95 Å, HAP-bound): the investigational comparator bersacapavir, the phase 2 heteroaryldihydropyrimidine GLS4 (morphothiadin), the discontinued vebicorvir, and the co-crystallised HAP_R01 as a re-dock control. Vebicorvir scored best at −8.03 kcal/mol, beating bersacapavir (−6.93 kcal/mol) by 1.10 kcal/mol, with HAP_R01 at −6.76 and GLS4 last at −6.53 kcal/mol — but the ranking is inverted against measured potency: vebicorvir is the weakest by experiment (EC50 173–250 nM), while GLS4 is the most potent (IC50 12 nM) and bersacapavir sits between them (EC50 45 nM). The 1.10 kcal/mol margin is inside Vina's ±1–2 kcal/mol noise, so the docking "win" is not a discovery. The falsifiable next step is a head-to-head HBV replication assay of vebicorvir against bersacapavir under identical conditions, with a pre-registered rule that a docking margin under 2 kcal/mol predicts no measurable potency advantage; a measured advantage for vebicorvir would refute the noise interpretation.

---

1. Introduction

Hepatitis B virus (HBV) remains one of the world's deadliest chronic infections: according to the WHO hepatitis B fact sheet, about 240 million people were living with chronic HBV infection in 2024, with about 0.9 million new infections each year, and hepatitis B caused an estimated 1.1 million deaths in 2024, mostly from cirrhosis and hepatocellular carcinoma. Pakistan carries a large share of that burden: WHO estimated about 3.8 million people living with hepatitis B in Pakistan in 2025, with only 25–30% aware of their infection, and a national survey found hepatitis B surface antigen (HBsAg) prevalence of 2.5% in the general population. Approved therapy — nucleos(t)ide analogues such as entecavir and tenofovir, and pegylated interferon — suppresses viral replication but almost never clears the virus, because none of these drugs eliminates covalently closed circular DNA (cccDNA) from the hepatocyte nucleus. Treatment is therefore usually lifelong, and functional cure — sustained HBsAg loss after stopping treatment — is the goal every new drug class is now judged against.

The HBV core protein (Cp, 183 amino acids) is the most advanced new drug target in this search. Cp self-assembles into the icosahedral capsid — predominantly T=4 particles of 120 dimers — that

packages pregenomic RNA (pgRNA) and the viral polymerase, and it acts at several steps of replication: assembly, pgRNA encapsidation, reverse transcription, nuclear trafficking and cccDNA formation. Capsid assembly modulators (CAMs), also called core protein allosteric modulators (CpAMs), bind a hydrophobic "HAP" pocket at the Cp dimer-dimer interface and misdirect assembly: class I CAMs (CAM-A, e.g. heteroaryldihydropyrimidines such as GLS4) induce aberrant, non-functional capsids, while class II CAMs (CAM-E, e.g. sulfamoylbenzamides such as bersacapavir) accelerate the formation of empty capsids devoid of pgRNA. Both classes prevent infectious virion formation, and both are being tested in combination regimens aimed at functional cure.

No CAM is approved. The most advanced, bersacapavir (JNJ-56136379 / JNJ-6379, a CAM-E), reached phase 2b: in the REEF-2 trial, 130 nucleos(t)ide analogue-suppressed, HBeAg-negative patients received 48 weeks of the siRNA JNJ-73763989 plus bersacapavir 250 mg daily plus a nucleos(t)ide analogue, followed by 48 weeks off treatment — and no patient achieved functional cure at follow-up week 24 or week 48, although mean HBsAg fell 1.89 log10 IU/mL on treatment versus 0.06 in the control arm. GLS4 (morphothiadin, a CAM-A heteroaryldihydropyrimidine) completed phase 2 in combination with ritonavir (which boosts its otherwise short half-life), and vebicorvir (a CAM-A) was discontinued after a phase 2b trial did not achieve the desired level of antiviral effect. The question this paper asks is methodological, in the same pattern as the DprE1 benchmark: can a standardised, automated docking pipeline rank known CAMs against bersacapavir, and what does an apparent docking "win" mean when the reference drug is investigational, the most potent compound by experiment scores worst, and the "winner" is a discontinued drug? The falsifiable hypothesis is that any CAM beating bersacapavir by less than 2 kcal/mol on Vina docking has no measurable potency advantage, and that the docking ranking will disagree with measured EC50 values on this target.

2. Methods

2.1 Literature basis

Sources were searched on 7 November 2026: the daily literature digest at `~/workspace/research-team/digests/`, PubMed and web search, ChEMBL (EBI), RCSB PDB and UniProtKB. Key sources: the WHO hepatitis B fact sheet for the global burden figures (240 million living with chronic HBV in 2024, 1.1 million deaths); the WHO Eastern Mediterranean Regional Office statement of 28 July 2025 for the Pakistan figure (about 3.8 million living with hepatitis B, 25–30% aware); the REEF-2 trial report (Journal of Hepatology 2024;81:404–414) for the functional-cure outcome; the 2023 Expert Opinion on Drug Discovery review of CAMs for the class definitions, trial status and resistance data; the 2013 Antimicrobial Agents and Chemotherapy characterisation of GLS4 (EC50 62.24 nM in HepAD38 cells, IC50 12 nM); and the 2021 PLoS Pathogens study of dimer-dimer interface residues for the HAP-pocket resistance mechanism. Measured potencies quoted in this paper are ChEMBL bioactivities against the organism-level target "Hepatitis B virus" (CHEMBL613497), retrieved live on 7 November 2026.

2.2 Molecular docking

Docking was performed with AutoDock Vina 1.2.7, using the LeadForge protocol: the same structure, grid centre, box (22.5 Å cube), exhaustiveness (8) and pose count (3) for every ligand; best of three poses reported. The structure was the HBV core protein, PDB 5WRE (1.95 Å, "Hepatitis B virus core protein Y132A mutant in complex with heteroaryldihydropyrimidine (HAP_R01)", ligand code 7TL), with the grid centred exactly on the centroid of the co-crystallised ligand 7TL, chain A: [–44.94, 9.56, 28.99]. The centre was verified against the PDB file to 0.01 Å: the pipeline's own ligand-centre function returns the same coordinates, and the nearest crystallisation additives (isopropyl alcohol, glycerol, chloride) sit 14–22 Å away, so the grid-centre fault that has corrupted earlier runs (centring

on a cofactor, glycan or solvent) did not occur here. The receptor was the full crystallographic hexamer (chains A–F), so the dimer-dimer interface pocket — which is formed by two dimers — is intact in the docking receptor. Receptor preparation used RDKit (Gasteiger charges, simplified AutoDock atom types); ligand preparation used meeko. Vina's scoring noise is approximately ±1–2 kcal/mol; differences smaller than this cannot be treated as significant on scores alone.

Four compounds were docked, all with identities verified in ChEMBL (canonical SMILES taken from ChEMBL, not from a text search): bersacapavir (CHEMBL4744254), GLS4 (CHEMBL3742114), vebicorvir (CHEMBL4650324), and HAP_R01 (CHEMBL4446995), the co-crystallised ligand of 5WRE, re-docked as a control. Three further candidates were considered and excluded because their identities could not be verified: ChEMBL text search returned no named entry for NVR 3-778, returned a non-matching structure for AT-130 (no bromine in the returned SMILES, while AT-130 contains bromine), and returned vebicorvir itself for ABI-H0731. They were not docked under those names.

2.3 PAINS screen

Every test compound was screened with the RDKit PAINS filter: all four compounds were PAINS-clean (no alerts). Lipinski descriptors were computed with RDKit: bersacapavir MW 418.37, logP 2.52, 0 violations; GLS4 MW 509.40, logP 3.29, 1 violation (MW > 500); vebicorvir MW 467.45, logP 3.49, 0 violations; HAP_R01 MW 514.91, logP 3.25, 1 violation (MW > 500).

2.4 Honesty limits

What this paper cannot show: docking is not activity; Vina noise is ±1–2 kcal/mol; no experimental data are presented; no molecular-dynamics refinement, binding-free-energy calculation, or re-docking RMSD validation was performed; a single crystal structure (5WRE, one core-protein conformation, a Y132A mutant) was used; and the structure's UniProt entry (L7R9I1) differs from the genotype D reference P03146 at six of 183 residues (96.7% identity), two of which (positions 35 and 40) sit in the pocket-adjacent 23–39 stretch. The comparator is investigational, not approved: there is no approved HBV capsid drug, so no approved reference exists. No cure is claimed and no clinical recommendation is made.

3. Results

3.1 Docking: the comparison table

Compound	Class	Measured potency, ChEMBL (HBV)	Best affinity (kcal/mol)	Poses (kcal/mol)	MW	logP	Lipinski violations	PAINS
Vebicorvir	CAM-A	EC50 173 nM; 250 nM	**–8.03**	–8.03, –7.93, –7.79	467.45	3.49	0	clean
Bersacapavir (comparator)	CAM-E	EC50 45 nM	–6.93	–6.93, –6.91, –6.89	418.37	2.52	0	clean
HAP_R01 (re-dock control)	CAM-A	— (co-crystallised ligand)	–6.76	–6.76, –6.39, –6.32	514.91	3.25	1	clean
GLS4 (morphothiadin)	CAM-A	IC50 12 nM; EC50 13 nM	–6.53	–6.53, –6.51, –6.43	509.40	3.29	1	clean

All docking on PDB 5WRE (HBV core protein Y132A mutant, 1.95 Å), grid centre [–44.94, 9.56, 28.99] (7TL chain-A centroid), box 22.5 Å, exhaustiveness 8, Vina 1.2.7, 7 November 2026. Measured potencies are ChEMBL bioactivities against target CHEMBL613497 ("Hepatitis B virus");

GLS4's published HepAD38 EC50 is 62.24 nM (Wu et al. 2013).

The headline, stated honestly: one genuine CAM "beats" bersacapavir on docking — vebicorvir by 1.10 kcal/mol (−8.03 versus −6.93) — but vebicorvir is the weakest of the three by measured potency (EC50 173–250 nM, 4–6 times weaker than bersacapavir's 45 nM and 14–21 times weaker than GLS4's 12 nM). The docking ranking is inverted against measured potency: the best-docking compound is the weakest by EC50, and the most potent compound (GLS4, 12 nM) scores worst on docking (−6.53). The margin (1.10 kcal/mol) is also inside Vina's ±1–2 kcal/mol noise, so it would not be interpretable as a real advantage even if the potency order agreed. Pose consistency does not rescue the ranking: vebicorvir holds its best score across all three poses (−8.03, −7.93, −7.79), yet is the weakest by EC50; GLS4 is equally pose-stable at the bottom of the table (−6.53, −6.51, −6.43). The re-dock control is informative in a different way: HAP_R01, re-docked into its own crystallographic pocket, scores −6.76 — in the same narrow band as bersacapavir and GLS4, and 1.27 kcal/mol worse than vebicorvir — which shows that a ligand crystallised in this exact pocket does not score best in it. The whole table spans only 1.50 kcal/mol across four compounds whose measured potencies span a 21-fold range.

3.2 Comparison with published docking

No published Vina score for bersacapavir, GLS4 or vebicorvir on 5WRE was found to benchmark against — which is itself a gap this paper fills: Vina 1.2.7 reference values for the three most clinically advanced CAMs on 5WRE are now on record for others to reproduce, together with the re-dock control. The ordering is not comparable to published potency: the three compounds span EC50/IC50 12–250 nM, a 21-fold spread, while their docking scores span 1.50 kcal/mol in the wrong order — Vina ranks poses, not EC50s, and on this target it ranks them backwards.

3.3 The standardisation / quality problem

Three things that could silently corrupt this result did not, and two pipeline faults were caught before reporting. The grid centre was verified against the PDB file to 0.01 Å and sits on the co-crystallised HAP ligand, 14–22 Å from the nearest solvent additive, so the cofactor/glycan-centring fault is absent here. The duplicate-entry fault seen in earlier papers did not occur: none of the four docked compounds is a duplicate of the comparator (all four have distinct molecular formulae and weights). But the automated pipeline failed twice before this manual run, and both failures are part of the result. First, the pipeline's UniProt lookup prefers human proteins, so the text query "Hepatitis B virus core protein" resolved to Q96SB4 — human SRSF protein kinase 1 (SRPK1), a kinase that phosphorylates the core protein but is not the capsid — and that run was killed before docking. Second, with the accession P03146 forced, UniProt resolved correctly (HBV genotype D capsid protein) but the pipeline's ChEMBL target fallback picked CHEMBL2074, human maltase-glucoamylase, and its 26 "leads" were maltase-glucoamylase ligands; those leads were discarded, not used. Third, the pipeline's PDB auto-pick found no ligand-bound X-ray structure under 3.0 Å cross-referenced to P03146, so docking was skipped — the structure used here, 5WRE, is cross-referenced to a different HBV core-protein entry (L7R9I1) and had to be selected manually. A pipeline that silently prefers human proteins, silently falls back to a wrong ChEMBL target, and cannot find a structure that exists under a sibling accession is a pipeline that needs its target-resolution step fixed before it is trusted on viral targets; the manual verification steps in this paper are the workaround, not the fix.

4. Discussion

4.1 The reframe

The result reframes the paper from a discovery claim into a measurement of Vina's limits on this target — and into a caution about the functional-cure literature. There is no lead here: the one

compound that "beats" bersacapavir on docking is the weakest by measured EC50, was discontinued after phase 2b for insufficient antiviral effect, and its margin sits inside Vina's noise. The most potent compound by experiment (GLS4, 12 nM, in phase 2 with ritonavir) scores worst. The durable findings are the benchmark values for the three clinically advanced CAMs on 5WRE, the re-dock control (–6.76 kcal/mol for a ligand in its own pocket), and the demonstration that on a capsid-assembly target, non-covalent docking ranks known CAMs in the opposite order to their measured potency — the same inversion the DprE1 benchmark found on a covalent-inhibitor target, which suggests the failure is general to this pipeline's scoring on assembly/modulator targets, not specific to one protein.

4.2 Why the "win" must not be over-read

Three reasons. First, the margin (1.10 kcal/mol) is inside Vina's ±1–2 kcal/mol noise. Second, the ranking is inverted against measured potency — the best-docking compound is the weakest by EC50, by a factor of 4–6 against the comparator and 14–21 against GLS4. Third, the scored quantity is a static binding energy at one pocket of a mutant hexamer, while CAM activity depends on the kinetics of capsid assembly, the distinction between aberrant and empty capsids, pgRNA packaging, and resistance — none of which a single-structure docking score can see. Bersacapavir monotherapy selected the T33N resistance mutation in the clinic, a reminder that the pocket itself evolves under drug pressure; a docking score computed on a wild-type-structure pocket says nothing about the resistant one. A docking "win" against an investigational comparator, on a mutant structure, is not a discovery; it is a measurement of what the method leaves out.

4.3 The PAINS problem is the experiment's design constraint, not a reason to stop

All four compounds are PAINS-clean, which removes one artefact class from the validation design — but PAINS-clean is a low bar, and two of the four carry a Lipinski violation (MW > 500), which is a developability note, not a disqualification: GLS4 is already in phase 2 despite it. The validation assay must still include a counter-screen and a capsid-assembly readout, because a clean PAINS filter does not rule out assay interference, and because a replication EC50 alone cannot distinguish a true CAM from a compound that merely poisons the cells.

4.4 Testable hypotheses

• Vebicorvir inhibits HBV replication with an EC50 within 2-fold of its ChEMBL values (173–250 nM) under identical assay conditions, and is less potent than bersacapavir (45 nM) — the docking predicts no measurable advantage; a measured EC50 for vebicorvir better than bersacapavir's would refute the noise interpretation.

• A Vina re-dock of the co-crystallised ligand 7TL (HAP_R01) into 5WRE reproduces its crystallographic pose within 2 Å RMSD (protocol self-validation; affinity only is reported here, RMSD not yet computed).

• Docking the same four compounds against a second HBV core-protein structure (for example a CAM-E-bound structure, if one becomes available) preserves the inversion against measured potency — testing whether the failure is structure-specific or method-general.

• Fixing the pipeline's target-resolution step (organism-aware UniProt lookup, accession-verified ChEMBL target, PDB search across sibling accessions) reproduces the manual 5WRE selection automatically — a pipeline fix to be verified, not assumed.

4.5 Proposed validation (university-lab feasible, 8–12 weeks)

Stage 1 (weeks 1–2): HBV replication assay in HepG2.2.15 or HepAD38 cells for vebicorvir, bersacapavir and GLS4, same plate, n = 3, HBV DNA by qPCR as readout, with a pre-registered 2-fold decision rule and the rule that a docking margin under 2 kcal/mol predicts no measurable advantage. Stage 2 (weeks 3–6): capsid-assembly readout (native agarose gel / capsid ELISA) for any compound that clears Stage 1, to confirm the mechanism is capsid modulation rather than

cytotoxicity, with a cell-viability counter-screen (CC50) on the same plate. Stage 3 (weeks 7–12): resistance check — sequence the core-protein HAP pocket (residues 23–40, 102–140) after serial passage under CAM pressure, looking for T33N and neighbouring substitutions, since pocket evolution is the known clinical failure mode for this class. Total compound cost is the binding constraint; all leads are investigational research chemicals, not approved drugs, and none may be presented as a treatment.

5. Limitations

• Docking only; no experimental data. 2. The single "winning" margin (1.10 kcal/mol) is inside Vina's ±1–2 kcal/mol noise and cannot be interpreted as a real advantage. 3. The docking ranking is inverted against measured potency, so the scores cannot be used to rank these CAMs. 4. No molecular-dynamics refinement, binding-free-energy calculation, or re-docking RMSD validation was performed. 5. A single crystal structure was used (5WRE, one core-protein conformation, a Y132A mutant whose UniProt entry differs from genotype D P03146 at six residues, two of them pocket-adjacent at positions 35 and 40); induced-fit effects and genotype variation are not modelled. 6. The comparator is investigational — no approved HBV capsid drug exists, so there is no approved reference standard. 7. All three CAMs are already-known compounds; no new chemical matter is proposed. 8. The automated pipeline's target resolution failed twice on this topic (wrong UniProt organism, wrong ChEMBL target) and its PDB auto-pick missed a structure that exists under a sibling accession — other silent defaults in the same pipeline are not yet audited. 9. No cccDNA or HBsAg-loss readout was modelled; docking says nothing about functional cure, which is a clinical endpoint, not a binding energy.

6. Conclusion

No CAM in this set beats bersacapavir: the one compound that scores better on Vina docking is the weakest by measured EC50 and was discontinued after phase 2b, the most potent compound by experiment scores worst, and the margin sits inside Vina's noise. The honest reading is that on a capsid-assembly target, non-covalent docking ranks known CAMs backwards — and the next experiment is a head-to-head HBV replication assay with a pre-registered 2 kcal/mol rule, plus a capsid-assembly readout and a resistance check at the HAP pocket, not more Vina docking.

Acknowledgments

This manuscript was prepared with AI assistance (Lapis) for drafting, literature search and docking computations. The author reviewed the manuscript and takes responsibility for its content. No funding was received. The author thanks the ChEMBL, UniProtKB, RCSB PDB and PubChem teams for free, open data.

References

• World Health Organization. Hepatitis B (fact sheet). Geneva: WHO; updated 2026. https://www.who.int/news-room/fact-sheets/detail/hepatitis-b (accessed 7 November 2026). (About 240 million people living with chronic hepatitis B in 2024; about 1.1 million deaths in 2024.)

• World Health Organization, Regional Office for the Eastern Mediterranean. WHO urges action to detect and treat over 13.8 million people infected with hepatitis B and C in Pakistan. 28 July 2025. https://www.emro.who.int/pak/pakistan-news/who-urges-action-to-detect-and-treat-over-13-8-million-people-infected-with-hepatitis-b-and-c-in-pakistan.html (accessed 7 November 2026).

• Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. *J Chem Inf Model.* 2021;61(8):3891–3898. PMID: 34278794.

• RCSB Protein Data Bank. Hepatitis B virus core protein Y132A mutant in complex with heteroaryldihydropyrimidine (HAP_R01). PDB ID 5WRE, 1.95 Å. https://www.rcsb.org/structure/5WRE (accessed 7 November 2026).

• UniProtKB. Capsid protein, Hepatitis B virus genotype D subtype ayw. UniProt P03146. https://www.uniprot.org/uniprotkb/P03146/entry (accessed 7 November 2026).

• ChEMBL database, European Bioinformatics Institute. Bersacapavir (CHEMBL4744254), GLS4/morphothiadin (CHEMBL3742114), vebicorvir (CHEMBL4650324), HAP_R01 (CHEMBL4446995); bioactivities against target CHEMBL613497 (Hepatitis B virus). https://www.ebi.ac.uk/chembl/ (accessed 7 November 2026).

• JNJ-73763989 and bersacapavir treatment in nucleos(t)ide analogue-suppressed patients with chronic hepatitis B: REEF-2. *J Hepatol.* 2024;81:404–414. DOI: 10.1016/j.jhep.2024.03.046. ClinicalTrials.gov NCT04129554.

• Wu G, et al. Preclinical characterization of GLS4, an inhibitor of hepatitis B virus core particle assembly. *Antimicrob Agents Chemother.* 2013;57(11):5344–5354. DOI: 10.1128/AAC.01091-13. PMID: 23959305.

• The premise of capsid assembly modulators towards eliminating HBV persistence. *Expert Opin Drug Discov.* 2023;18(9):1031–1041. DOI: 10.1080/17460441.2023.2239701. PMCID: PMC10530454.

• Core protein: a pleiotropic keystone in the HBV lifecycle. *Antiviral Res.* 2015;121:82–93. DOI: 10.1016/j.antiviral.2015.06.020. PMID: 26129969.

• Liu H, Cheng J, Viswanathan U, Chang J, Lu F, Guo J-T. Amino acid residues at core protein dimer-dimer interface modulate multiple steps of hepatitis B virus replication and HBeAg biogenesis. *PLoS Pathog.* 2021;17(11):e1010057. DOI: 10.1371/journal.ppat.1010057.

• ClinicalTrials.gov. Safety, antiviral activity, and pharmacokinetics of morphothiadine mesilate capsules / ritonavir tablets in chronic hepatitis B (phase 2). NCT03638076. https://clinicaltrials.gov/study/NCT03638076 (accessed 7 November 2026).

Data availability

All docking scores reported here are in `hidden_files/dock-HP-059-2026-11-07c.json` (Vina 1.2.7, PDB 5WRE, centre [–44.94, 9.56, 28.99]). The pipeline run record is in `hidden_files/dock-HP-059-2026-11-07.json`; its lead list was discarded as wrong-target (human maltase-glucoamylase ligands) and is not used in this paper — see §3.3. ChEMBL IDs: bersacapavir CHEMBL4744254, GLS4 CHEMBL3742114, vebicorvir CHEMBL4650324, HAP_R01 CHEMBL4446995. Target: UniProt P03146 (HBV genotype D capsid protein); structure UniProt L7R9I1 (5WRE). PDB ID 5WRE. No additional data were generated.